

Yes. We can hold Incident 04 exactly where it is and now expand **Incident 05 as the complete bridge between “we were compromised” and “we are ready to conduct a recovery operation.”**

Incident 05 is structurally important because this is where the learner finally understands **why networking evidence matters and why asymmetric cryptography is needed.**

The learner is **not attacking anything yet.** Incident 05 prepares a trustworthy recovery command architecture so that Incident 06 can contain investigation, Red Team action, evidence collection and recovery without Phantom Q blindly trusting messages again.

Incident 05 — Rebuild Trust for the Recovery Operation

1. Where we are coming from

Incident 04 ended with:

PQ-004

INTENDED

R100 → •••• 4821

ACTUAL

R1 000 → •••• 7394

KEY A7

COMPROMISED

COMMUNICATION LINK

SUSPENDED

NETWORK SOURCE RECORD

172.18.44.7

ACTOR

UNKNOWN

R1 000

NOT RECOVERED

The most important lesson from Incident 04 was:

MESSAGE DECRYPTS WITH A7

≠

ALICE DEFINITELY CREATED IT

Therefore Phantom Q now has two problems.

PROBLEM 1

We have evidence pointing somewhere on the network.

PROBLEM 2

We cannot safely coordinate the recovery team using the old shared-secret trust model.

Incident 05 solves **both enough for Release 1**.

2. The core purpose of Incident 05

Incident 05 asks four questions in sequence:

1. WHAT IS 172.18.44.7?
2. DOES THAT IP ADDRESS IDENTIFY THE ATTACKER?
3. HOW CAN HQ SEND INFORMATION
TO RED TEAM WITHOUT REUSING A7?
4. HOW CAN RED TEAM PROVE
THAT A COMMAND REALLY CAME FROM HQ?

Then, once that trust is established:

5. HOW DO HQ AND RED TEAM
CREATE A FRESH OPERATIONAL SESSION?

The answer eventually becomes:

```
ASYMMETRIC SECURITY
  ↓
ESTABLISH / VERIFY TRUST
  ↓
SESSION KEY S1
  ↓
SYMMETRIC SECURITY
  ↓
LIVE RECOVERY COMMUNICATION
```

That architecture becomes operational in Incident 06.

3. What Incident 05 does NOT do

We must protect the scope.

Incident 05 does **not** teach:

- subnetting;
- CIDR;
- routing tables;
- NAT;
- packet analysis;
- full RSA mathematics;
- certificates/PKI in depth;
- advanced signature algorithms;
- penetration testing;
- exploitation;
- attribution;
- money recovery.

And importantly:

```
172.18.44.7
≠
ATTACKER IDENTITY
```

It is a **network lead**.

4. Starting state

The learner begins in an escalated Phantom Q environment.

```
SECURITY STATUS
RESTRICTED

PQ-004
ACTIVE

R1 000
MISSING

A7
COMPROMISED / RETIRED

OLD LINK
SUSPENDED

EVIDENCE
```

PRESERVED

NETWORK RECORD

172.18.44.7

RED TEAM

STANDBY

The mission objective appears:

REBUILD TRUST FOR THE RECOVERY OPERATION

5. Incident 05 — Locked 28-scene structure

The authoritative sequence is:

```
01 Recovery Mission Opens
02 Find Suspicious Session
03 What Is an IP Address?
04 Mark IP as Lead
05 Red Team Goes on Standby
06 A7 Cannot Be Reused
07 Open Asymmetric Workbench
08 Create Public Component
09 Generate Key Pair
10 Public Can Leave — Private Cannot
11 Red Team Builds Its Key Pair
12 Register Red Team Public Key
13 Create Recovery Brief
14 Protect It for Red Team
15 Send Through Existing Network
16 What Can the Network Still See?
17 Red Team Uses Its Private Key
18 But Who Sent It?
19 HQ Builds Its Key Pair
20 HQ Signs Recovery Instruction
21 Red Team Verifies HQ
22 Tamper Test
23 Build Trusted Command Flow
24 Create Fresh Session Key S1
25 Deliver S1 Using Asymmetric Protection
26 Open Operational Session
27 Secure Does Not Mean Invisible
28 Incident 05 End State — Red Team Recovery Operation Ready
```

Incident 05 is not complete before Scene 28.

PART A — TURN PQ-004 INTO A NETWORK LEAD

Scenes 1–6

Scene 1 — Recovery Mission Opens

The learner returns to the PQ-004 case.

The screen presents the unresolved impact:

PQ-004 RECOVERY

Financial impact

R1 000

Recovery status

NOT RECOVERED

A7

COMPROMISED

Actor

UNKNOWN

HQ states:

The incident has been contained, but the diverted funds remain unrecovered.

New mission:

PREPARE THE RECOVERY OPERATION

The learner does not immediately get a “Trace Hacker” button.

First they reopen the evidence.

Scene 2 — Find the Suspicious Session

The player examines preserved Incident 04 session evidence.

Expected activity:

A7 ACTIVITY

Alice

EXPECTED

Bob

EXPECTED

Unknown session

UNAUTHORISED

Selecting the unknown session reveals:

SESSION RECORD

Case:

PQ-004

Time:

14:32:18

Source:

172.18.44.7

Status:

UNAUTHORISED / SUSPICIOUS

The learner now has a technical clue.

But most beginners may not yet know what:

172.18.44.7

means.

That naturally creates Scene 3.

Scene 3 — What Is an IP Address?

This should be extremely simple and visual.

We show devices/systems on a network.

SYSTEM A

10.14.2.11

SYSTEM B

172.18.44.7

At this level:

An IP address is a network address used to identify a point/device/interface involved in communication.

For the current mission, we care about:

SOURCE

Where traffic appears to originate

DESTINATION

Where traffic is going

The learner can place:

172.18.44.7

into:

SOURCE

No subnetting.

No binary conversion.

No address classes.

Scene 4 — Mark the IP as a Lead

Now the system asks:

What can we conclude from 172.18.44.7?

Possible claims might include:

- A. This is definitely the attacker.
- B. This is a network point associated with suspicious activity.
- C. This identifies the physical person responsible.

Correct evidence-based conclusion:

172.18.44.7

CLASSIFICATION

NETWORK LEAD ✓

ACTOR IDENTITY

UNKNOWN

The key rule appears:

IP ADDRESS ≠ PERSON

This carries directly into Incident 06.

The learner records:

NETWORK LEAD
172.18.44.7

in the PQ-004 evidence board.

Scene 5 — Red Team Goes on Standby

Now Phantom Q determines that the clue may require specialist investigation.

The Red Team Operations area changes from:

LOCKED

to:

RED TEAM
STANDBY

Not yet deployed.

HQ prepares to send:

- case information;
- network lead;
- financial lead;
- operational instructions.

But before sending anything, the system checks the current secure channel.

Scene 6 — A7 Cannot Be Reused

A7 appears in the trust system:

KEY A7

STATUS
COMPROMISED

TRUST
REVOKED

USE

PROHIBITED

The learner attempts:

SEND RECOVERY ORDER USING A7

Blocked:

TRUST FAILURE

A compromised shared secret
cannot be used to establish
trusted recovery communication.

This is where Incident 05's true cryptographic problem appears.

6. The conceptual bridge

At this point the learner understands:

WE NEED TO COMMUNICATE
WITH RED TEAM

BUT

WE CANNOT REUSE A7

AND

A SHARED SECRET ALONE
DID NOT PROVE WHO CREATED A MESSAGE

So Phantom Q needs another security capability.

That unlocks:

ASYMMETRIC WORKBENCH

PART B — INTRODUCE ASYMMETRIC SECURITY

Scenes 7–12

Scene 7 — Open the Asymmetric Workbench

The learner enters the Cryptography Lab.

Previously used modules remain visible:

PLAINTEXT
CIPHERTEXT
A7
ENCRYPT
DECRYPT

A new section activates:

ASYMMETRIC SECURITY

PUBLIC
PRIVATE

The player is not immediately given a wall of theory.

The system simply says:

We need a security model where one key can be shared while another remains secret.

Scene 8 — Create the Public Component

We use deliberately small **training numbers**.

11
×
13
=
143

The player physically combines:

11
13

to create:

143

The game labels it something like:

PUBLIC COMPONENT — 143

Important:

This is not a full RSA mathematics lesson.

The purpose is only to expose the learner to:

```
private inputs
↓
mathematical relationship
↓
public component
```

The training system handles the remaining key-generation mathematics.

Scene 9 — Generate the Key Pair

The player selects:

GENERATE TRAINING KEY PAIR

Result:

PUBLIC KEY

Generated ✓

PRIVATE KEY

Generated ✓

They appear as two physically distinct objects.

```
      KEY PAIR
     /      \
PUBLIC      PRIVATE
```

Scene 10 — Public Can Leave, Private Cannot

Now the learner experiments.

Public key

Drag toward shared/network area:

PUBLIC KEY

↓

SHARE

Private key

Try to drag outward:

PRIVATE KEY

↓

SHARE

System:

BLOCKED

PRIVATE KEY

MUST REMAIN WITH OWNER

This gives the basic rule:

PUBLIC KEY

Can be distributed.

PRIVATE KEY

Must remain under owner's control.

That is enough for Act I.

Scene 11 — Red Team Builds Its Key Pair

Now we apply the concept to an actual role.

Inside Red Team Operations:

RED TEAM UNIT 01

generates:

RT PUBLIC KEY

RT PRIVATE KEY

The critical visual relationship:

RED TEAM UNIT 01

PRIVATE KEY

STAYS LOCAL ✓

```
PUBLIC KEY
AVAILABLE TO REGISTER ✓
```

The Red Team private key should never magically appear on HQ screens.

Scene 12 — Register the Red Team Public Key

The Red Team sends/registers only:

```
RT PUBLIC KEY
```

HQ records:

```
TRUST REGISTRY

OWNER
Red Team Unit 01

PUBLIC KEY
REGISTERED ✓
```

Now HQ has something it can safely use to protect information specifically for Red Team.

This is the first real operational application of asymmetric security.

PART C — PROTECT THE RECOVERY BRIEF

Scenes 13–18

Scene 13 — Create the Recovery Brief

HQ constructs the first PQ-004 recovery brief.

Locked core content:

```
PQ-004 RECOVERY BRIEF

Financial impact:
R1 000

Financial lead:
.... 7394

Network lead:
```

172.18.44.7

Actor:

UNKNOWN

INSTRUCTION:

Review evidence.

Do not engage yet.

This last line matters:

DO NOT ENGAGE YET

Incident 05 is preparing Red Team, not authorising an attack.

Scene 14 — Protect It for Red Team

HQ selects:

RED TEAM PUBLIC KEY

The learner assembles:

RECOVERY BRIEF

+

RED TEAM PUBLIC KEY

↓

PROTECT

↓

PROTECTED RECOVERY BRIEF

The important conceptual relationship:

Information protected for Red Team using the Red Team public key is recoverable using the corresponding Red Team private key.

The learner does not need deeper maths.

Scene 15 — Send Through the Existing Network

Now something interesting happens.

The protected recovery brief still has to use the same physical/network infrastructure.

HQ

↓

PROTECTED BRIEF

↓

NETWORK

↓

RED TEAM

The learner might ask:

But we already said the network cannot simply be trusted.

Exactly.

Cryptography should allow secure communication **over an untrusted or observable path**, rather than requiring the path itself to become magical.

The protected brief is sent.

Scene 16 — What Can the Network Still See?

We return to the communication-channel lesson from Incident 04.

A network observation display shows:

SOURCE

HQ network point

DESTINATION

Red Team network point

TIME

Visible

TRAFFIC EXISTS

Visible

PROTECTED CONTENT

Visible as ciphertext/protected data

But:

RECOVERY BRIEF PLAINTEXT

Not directly visible from passive observation

This reinforces:

SECURE CONTENT

≠

Very important for Scene 27 later.

Scene 17 — Red Team Uses Its Private Key

Red Team receives the protected brief.

They use:

```
RT PRIVATE KEY
```

to recover it.

```
PROTECTED BRIEF
+
RT PRIVATE KEY
↓
RECOVER
↓
PQ-004 RECOVERY BRIEF
```

Result:

```
R1 000
7394
172.18.44.7
Actor unknown
Do not engage yet
```

The learner sees that Red Team successfully obtained the intended content.

At first this appears to solve the problem.

Then comes the most important question.

Scene 18 — But Who Sent It?

Red Team asks:

“We can read the recovery brief. But how do we know HQ actually sent it?”

This must be a major narrative beat.

Because Incident 04 proved:

MESSAGE CLAIMS TO BE FROM ALICE
≠
PROOF ALICE CREATED IT

Now the same principle is applied at a higher operational level.

Imagine the consequence if Red Team receives:

ATTACK 172.18.44.7

and simply trusts the text:

FROM: HQ

That is unacceptable.

So encryption/decryption has solved:

WHO CAN READ IT?

but not yet fully solved:

WHO CREATED / AUTHORISED IT?

That creates digital signatures naturally.

PART D — PROVE HQ ORIGIN

Scenes 19–23

Scene 19 — HQ Builds Its Key Pair

HQ now creates its own:

HQ PUBLIC KEY
HQ PRIVATE KEY

Relationship:

HQ PRIVATE KEY
STAYS WITH HQ

HQ PUBLIC KEY
AVAILABLE TO RED TEAM

This key pair has a different operational purpose from Red Team's confidentiality pair, even though the same asymmetric principle underlies both.

Scene 20 — HQ Signs the Recovery Instruction

HQ prepares an instruction:

```
PQ-004 RECOVERY INSTRUCTION
```

```
Review network lead:  
172.18.44.7
```

```
Engagement:  
NOT YET AUTHORISED
```

HQ then uses:

```
HQ PRIVATE KEY
```

to create a signature.

Simplified visual:

```
RECOVERY INSTRUCTION  
    +  
HQ PRIVATE KEY  
    ↓  
SIGN  
    ↓  
SIGNED INSTRUCTION
```

The learner should understand:

HQ keeps the private key under its control, so a valid signature provides evidence tied to HQ's signing capability.

We do not need certificate-chain theory.

Scene 21 — Red Team Verifies HQ

Red Team receives:

```
SIGNED RECOVERY INSTRUCTION
```

and uses:

```
HQ PUBLIC KEY
```

to verify.

Result:

```
SIGNATURE  
VALID ✓  
  
CLAIMED ORIGIN  
HQ  
  
VERIFIED ORIGIN  
HQ ✓  
  
MESSAGE CHANGED  
NO ✓
```

Now the learner has solved the Incident 04 origin problem in a new way.

The conceptual contrast should be explicit:

```
INCIDENT 04  
  
A7 valid  
↓  
proved shared-secret possession  
↓  
did NOT prove Alice specifically created message  
  
INCIDENT 05  
  
HQ signature valid  
↓  
provides verifiable HQ origin  
+  
message integrity
```

That is one of the biggest learning payoffs in Release 1.

Scene 22 — Tamper Test

We now run an authorised training replay.

Original signed message:

DO NOT ENGAGE

The training system modifies it to:

ENGAGE NOW

Then Red Team attempts verification.

Result:

SIGNATURE VALID?

NO ✕

MESSAGE STATE

ALTERED

This gives the learner direct evidence that a signature supports:

ORIGIN

+

INTEGRITY

This is much more powerful than just defining digital signatures.

Scene 23 — Build the Trusted Command Flow

Now the learner must construct the new command architecture.

They receive blocks:

HQ CREATES COMMAND

HQ SIGNS

SEND

RED TEAM VERIFIES

VALID?

ACCEPT / REJECT

Correct flow:

HQ CREATES COMMAND

↓

HQ SIGNS WITH PRIVATE KEY

↓

SEND

↓

RED TEAM VERIFIES
WITH HQ PUBLIC KEY

↓

VALID?

✓

YES

↓

ACCEPT

✗

NO

↓

REJECT

This is the new **trusted-command flow**.

Now Phantom Q can prove who authorised sensitive instructions.

But we still do not want to use asymmetric operations for every routine mission message.

That creates the next stage.

PART E — BUILD A FRESH OPERATIONAL SESSION

Scenes 24–28

Scene 24 — Create Fresh Session Key S1

Phantom Q generates:

SESSION KEY S1

This must visually be different from A7.

A7

COMPROMISED / RETIRED ✗

S1

NEW / FRESH ✓

The learner should understand:

S1 IS NOT A7 REACTIVATED

It is a completely fresh secret for the recovery operation.

Why return to symmetric security?

This is important.

We do **not** teach:

asymmetric good, symmetric bad.

Instead:

ASYMMETRIC

Excellent for establishing/verifying trust
and safely setting up secrets

SYMMETRIC

Efficient and suitable for ongoing
session communication

So the architecture becomes:

ASYMMETRIC

↓

ESTABLISH TRUST

↓

SESSION KEY S1

↓

SYMMETRIC

↓

ROUTINE OPERATIONAL TRAFFIC

This prepares exactly how Incident 06 operates.

Scene 25 — Deliver S1 Using Asymmetric Protection

HQ needs Red Team to obtain S1 safely.

It uses Red Team's registered public key.

SESSION KEY S1

+

RT PUBLIC KEY

↓

PROTECT

↓

PROTECTED S1

↓

NETWORK

↓

RED TEAM

+

RT PRIVATE KEY

↓
RECOVER S1

Result:

HQ
S1 ✓

RED TEAM
S1 ✓

Now both authorised parties possess the fresh shared session secret.

Scene 26 — Open Operational Session

HQ and Red Team initiate the mission communication channel using:

SESSION S1

The system performs a simple test.

HQ:

RECOVERY CHANNEL TEST

Protected with S1.

Red Team recovers it.

Red Team sends response.

RED TEAM READY

HQ recovers it.

Result:

SESSION S1

HQ ↔ RED TEAM

STATUS
OPERATIONAL ✓

Routine recovery communication can now use S1.

Scene 27 — Secure Does Not Mean Invisible

This scene is essential and must not be skipped.

The learner observes the newly secured HQ ↔ Red Team session from the network perspective.

They may still see:

SOURCE

Visible

DESTINATION

Visible

TIME

Visible

TRAFFIC

Visible

SESSION ACTIVITY

Observable

But not necessarily:

PLAINTEXT MISSION CONTENT

Therefore:

SECURE ≠ INVISIBLE

And importantly:

172.18.44.7

REMAINS A NETWORK LEAD

NOT

A PROVEN HUMAN IDENTITY

This prevents the learner from leaving Incident 05 with an exaggerated idea that encryption makes network activity disappear.

Scene 28 — Incident 05 End State

Only now is Incident 05 complete.

The status board becomes:

INCIDENT 05

PQ-004

ACTIVE RECOVERY CASE

R1 000

NOT RECOVERED

NETWORK LEAD

172.18.44.7 ✓

ACTOR IDENTITY

UNKNOWN

A7

RETIRED / UNTRUSTED

RED TEAM PUBLIC KEY

REGISTERED ✓

RED TEAM PRIVATE KEY

LOCAL / PROTECTED ✓

HQ KEY PAIR

ACTIVE ✓

HQ SIGNATURE VERIFICATION

AVAILABLE ✓

TRUSTED COMMAND FLOW

OPERATIONAL ✓

SESSION KEY S1

ACTIVE ✓

HQ ↔ RED TEAM SESSION

READY ✓

RED TEAM

RECOVERY OPERATION READY

Then:

INCIDENT 05 COMPLETE

And the next objective:

BEGIN RECOVERY OPERATION

This is the only appropriate point to hand off into Incident 06.

7. What the player physically does

Across Incident 05, the learner should:

1. reopen PQ-004;
2. inspect the suspicious A7 session;
3. identify `172.18.44.7`;
4. learn source/destination IP basics;
5. classify the address as a network lead;
6. activate Red Team standby;
7. attempt/consider A7 and see why it cannot be reused;
8. enter the asymmetric workbench;
9. combine 11 and 13;
10. create public component 143;
11. generate a training key pair;
12. test public vs private key behaviour;
13. have Red Team generate its pair;
14. register Red Team's public key;
15. construct the recovery brief;
16. protect the brief for Red Team;
17. send it across the network;
18. inspect what passive network observation still reveals;
19. have Red Team recover the brief with its private key;
20. encounter the "who actually sent it?" problem;
21. generate HQ's key pair;
22. sign the recovery instruction;
23. verify HQ's signature;
24. run the tamper test;
25. construct trusted-command logic;
26. generate fresh Session Key S1;
27. protect S1 for Red Team;
28. recover S1 at Red Team;
29. establish the operational S1 session;
30. test HQ ↔ Red Team communication;
31. observe session metadata;
32. complete the final readiness assessment.

That gives Incident 05 plenty of interaction without turning into a cryptography lecture.

8. Information release

The learner should discover the incident progressively.

Beginning

We have a suspicious technical record:
172.18.44.7.

After networking section

It is a network lead.
It does not identify the person responsible.

After A7 review

We cannot use our previous shared secret.

After first asymmetric exercise

Public and private keys do different jobs.
The public key can be shared.
The private key must remain controlled.

After protecting Red Team's brief

We can protect information for a particular private-key holder.

After Scene 18

Being able to decrypt a message
does not itself prove HQ created it.

After signature verification

We can verify HQ origin and detect message alteration.

After S1

We can use asymmetric trust
to safely establish a fresh symmetric session.

End

HQ and Red Team can now communicate operationally.

The recovery operation can begin.

The actor is still unknown.

The R1 000 is still missing.

Perfect boundary into Incident 06.

9. Character roles

Trainee

The trainee develops significantly here.

They move from:

“A suspicious IP appeared.”

to:

“I understand that an IP is evidence rather than identity, and I can establish a verifiable trusted command channel for the recovery team.”

That's a major competence step.

Alice

Alice becomes less central during Incident 05.

That is appropriate.

Her source evidence remains inside PQ-004, but this incident is now mainly about:

HQ
↔
RED TEAM

She does not need to be artificially inserted into every scene.

Bob

Same principle.

Bob remains part of the PQ-004 evidence history, but he is not the central operator during the recovery preparation.

Again, we should not force him into every room.

Eve

Eve remains authorised security support.

She could help contextualise:

NETWORK OBSERVATION
A7 STATUS
SECURITY MONITORING

But she must not suddenly become linked to:

172.18.44.7

or be visually implied to be the actor.

Red Team

This is Red Team's formal introduction as an operational entity.

Progression:

EARLIER INCIDENTS

Restricted / unavailable

INCIDENT 05

Standby

↓

Create identity/key material

↓

Establish trust

↓

Establish S1

↓

READY

INCIDENT 06

DEPLOY

This makes their activation feel earned.

HQ

Incident 05 also makes **HQ itself a security identity**.

HQ is no longer just the building or narrator.

It gains:

HQ PRIVATE KEY

HQ PUBLIC KEY

SIGNED AUTHORITY

That becomes extremely important in Incident 06 when HQ authorises offensive action and financial recovery.

10. Persistent security architecture after Incident 05

By the end, the campaign has evolved from this:

INCIDENT 03

ALICE

A7

↓

A7

BOB

to:

INCIDENT 05

```
graph TD
    HQ[HQ] --> HQ_PKEY[HQ PRIVATE KEY]
    HQ_PKEY --> HQ_PKEY_PUB[HQ PUBLIC KEY]
    HQ_PKEY_PUB --> SA[SIGN AUTHORITY]
    SA --> RT[RED TEAM]
    RT --> RT_PKEY_LOCAL[RT PRIVATE KEY LOCAL]
    RT_PKEY_LOCAL --> RT_PKEY_REGISTERED[RT PUBLIC KEY REGISTERED]
    RT_PKEY_REGISTERED --> S1[ESTABLISH SESSION S1]
    S1 --> S1_LINK[HQ ← S1 → RED TEAM]
```

This architectural progression is one of the strongest pieces of the whole campaign.

11. Failure logic

We should again use consequences, not “wrong”.

Player says 172.18.44.7 is the attacker

ATTRIBUTION NOT SUPPORTED

The IP identifies a network point
associated with suspicious activity.

Human identity has not been established.

Try to reuse A7

TRUST FAILURE

A7 is recorded as compromised.

Recovery communication cannot rely
on a retired shared secret.

Share private key

PRIVATE KEY EXPOSURE

The owner's private key must remain
under the owner's control.

Action blocked.

Protect recovery brief using HQ public key instead of Red Team's when goal is Red Team confidentiality

The simulation should show that the selected key relationship does not satisfy the intended recipient-controlled recovery flow.

It should explain the relationship, not simply say “wrong key”.

Red Team trusts unsigned command

ORIGIN UNVERIFIED

The message may be readable,
but HQ authority has not been proven.

Accept tampered command

SIGNATURE VERIFICATION FAILED

The instruction differs from

the signed state.

Operational action blocked.

Reactivate A7 instead of creating S1

COMPROMISED SECRET

A fresh operational session
requires new key material.

12. Real concepts underneath the gameplay

Gameplay	Real concept
Inspect <code>172.18.44.7</code>	IP/network evidence
Mark as network lead	Attribution discipline
A7 blocked	Key compromise/retirement
$11 \times 13 \rightarrow 143$	Simplified asymmetric maths
Generate two keys	Public/private key pair
Share public, retain private	Asymmetric key ownership
RT key pair	Recipient-specific cryptographic identity
Protect recovery brief	Asymmetric confidentiality concept
Observe network metadata	Encryption doesn't hide all metadata
RT private-key recovery	Private-key control
"Who sent it?"	Authentication/origin problem
HQ signature	Digital signature
Verify with HQ public key	Origin/integrity verification
Tamper test	Signature integrity detection

Gameplay	Real concept
S1	Fresh symmetric session key
Protect S1 asymmetrically	Hybrid cryptographic architecture
Use S1 operationally	Efficient symmetric communication
Secure ≠ invisible	Network-security realism

13. What Incident 05 deliberately leaves unresolved

At Scene 28:

WHO OPERATED 172.18.44.7?

UNKNOWN

IS 172.18.44.7 THE FINAL SYSTEM?

UNKNOWN

WHERE DID THE NETWORK TRAIL GO NEXT?

UNKNOWN

WHERE IS THE R1 000 NOW?

UNKNOWN

WHO IS THE ACTOR?

UNKNOWN

What **has** been solved is:

CAN HQ TRUST RED TEAM COMMUNICATION?

YES

CAN RED TEAM VERIFY HQ AUTHORITY?

YES

CAN A FRESH OPERATIONAL SESSION BE ESTABLISHED?

YES

IS RED TEAM READY TO INVESTIGATE?

YES

That distinction is fundamental.

14. Persistent outputs created

Incident 05 creates:

```
PQ-004 RECOVERY CASE
|
|— Network Lead
|   172.18.44.7
|
|— Red Team Identity
|   Unit 01
|
|— RT Public Key
|   Registered
|
|— RT Private Key
|   Local only
|
|— HQ Public Key
|
|— HQ Private Key
|
|— Signed Command Method
|
|— Signature Verification Method
|
|— Session Key S1
|
|— Trusted HQ ↔ Red Team Session
```

All of these should remain available in Incident 06.

We should not make the learner “relearn” them.

Incident 06 should **use them operationally**.

15. Incident 05 success condition

The learner succeeds only when:

```
NETWORK LEAD
172.18.44.7 ✓

IP ≠ HUMAN IDENTITY
UNDERSTOOD ✓

A7
RETIRED ✓
```

RED TEAM KEY PAIR
CREATED ✓

RT PUBLIC KEY
REGISTERED ✓

RT PRIVATE KEY
RETAINED LOCALLY ✓

RECOVERY BRIEF
PROTECTED ✓

RED TEAM
CAN RECOVER BRIEF ✓

HQ KEY PAIR
ACTIVE ✓

HQ SIGNATURE
VERIFIABLE ✓

TAMPERING
DETECTABLE ✓

TRUSTED COMMAND FLOW
COMPLETE ✓

SESSION KEY S1
CREATED ✓

S1 DELIVERY
SECURELY ESTABLISHED ✓

HQ ↔ RED TEAM
OPERATIONAL ✓

R1 000
STILL NOT RECOVERED

ACTOR
STILL UNKNOWN

The last two are not failures.

They are why Incident 06 exists.

16. Incident 05 closing sequence

The final moments could work like this:

Red Team Operations comes online.

Their central screen shows:

RED TEAM UNIT 01

HQ AUTHORITY

VERIFIED ✓

SESSION S1

ACTIVE ✓

RECOVERY CASE

PQ-004

NETWORK LEAD

172.18.44.7

FINANCIAL LEAD

•••• 7394

FINANCIAL LOSS

R1 000

ACTOR

UNKNOWN

STATUS

READY

HQ sends one final signed instruction:

RECOVERY OPERATION

PREPARE TO DEPLOY

Red Team verifies it.

HQ SIGNATURE

VALID ✓

Then:

INCIDENT 05 COMPLETE

RED TEAM RECOVERY OPERATION READY

And only then do we enter Incident 06.

Incident 05 master flow

```
INCIDENT 04
R1 000 MISSING
↓
A7 COMPROMISED
↓
OPEN PQ-004 RECOVERY CASE
↓
FIND UNKNOWN SESSION
↓
172.18.44.7
↓
WHAT IS AN IP?
↓
SOURCE / DESTINATION BASICS
↓
IP ≠ PERSON
↓
MARK 172.18.44.7
AS NETWORK LEAD
↓
RED TEAM STANDBY
↓
NEED TRUSTED RECOVERY COMMUNICATION
↓
TRY A7
↓
BLOCKED — COMPROMISED
↓
OPEN ASYMMETRIC WORKBENCH
↓
11 × 13 = 143
↓
GENERATE PUBLIC / PRIVATE KEY PAIR
↓
PUBLIC CAN BE SHARED
PRIVATE STAYS WITH OWNER
↓
RED TEAM GENERATES KEY PAIR
↓
REGISTER RT PUBLIC KEY
↓
CREATE PQ-004 RECOVERY BRIEF
↓
PROTECT USING RT PUBLIC KEY
↓
SEND THROUGH NETWORK
↓
NETWORK STILL SEES METADATA / TRAFFIC
↓
```

RED TEAM USES RT PRIVATE KEY
↓
RECOVERY BRIEF OPENED
↓
BUT WHO SENT IT?
↓
HQ GENERATES KEY PAIR
↓
HQ SIGNS RECOVERY INSTRUCTION
↓
RED TEAM USES HQ PUBLIC KEY
↓
HQ ORIGIN VERIFIED
↓
TAMPER TEST
↓
ALTERED MESSAGE FAILS VERIFICATION
↓
BUILD TRUSTED COMMAND FLOW
↓
CREATE FRESH SESSION KEY S1
↓
PROTECT S1 FOR RED TEAM
↓
RED TEAM RECOVERS S1
↓
HQ S1 = RED TEAM S1
↓
OPEN OPERATIONAL SESSION
↓
TEST COMMUNICATION
↓
SECURE ≠ INVISIBLE
↓
172.18.44.7 REMAINS LEAD
NOT IDENTITY
↓
R1 000 STILL MISSING
↓
ACTOR STILL UNKNOWN
↓
RED TEAM READY
↓
INCIDENT 05 COMPLETE
↓
INCIDENT 06
RECOVERY OPERATION

This is the full **Incident 05 design specification**. Its real achievement is not “teaching asymmetric cryptography” in isolation. It makes the learner understand **why asymmetric security becomes necessary after Incident 04**, then combines it with a fresh symmetric

Session Key S1 so Phantom Q has a technically coherent command architecture ready for the recovery operation.